

# ICN- in-network computing for UAS operator - UAS network slice

In/network computing for SLAM, Point cloud, UTM, inter- intra CNS network

In-network computing can significantly enhance distributed communication for **Unmanned Traffic Management (UTM)** systems, particularly within frameworks like **U-Space**, which is Europe's approach to integrating drones into the airspace. Here's how:

## 1. Reducing Latency in Decision-Making

- U-Space services involve real-time or near-real-time decision-making for drone operations, including conflict detection, dynamic re-routing, and separation management.
- By using in-network computing (processing data within network nodes rather than in centralized cloud systems), UTM systems can **minimize communication delays**, ensuring faster responses to airspace changes.

## 2. Optimizing Data Flow in a Distributed Architecture

- In traditional UTM implementations, data from multiple stakeholders (e.g., drones, operators, regulators) is sent to cloud servers for processing.
- In-network computing **offloads computations to edge nodes**, reducing congestion and enhancing bandwidth efficiency.
- This is critical in environments with **high-density drone operations**, where network bottlenecks could cause delays in critical alerts.

## 3. Enhancing Scalability for Large-Scale Drone Operations

- U-Space envisions a future with thousands of simultaneous drone flights.
- In-network computing allows **localized processing at the network edge**, ensuring scalability without overloading central systems.

## 4. Improving Security and Data Privacy

- Sensitive flight data and navigation information can be processed closer to the source, **reducing exposure to cyber threats** associated with transmitting data to centralized systems.
- Distributed security mechanisms can be embedded within the network fabric to detect anomalies and prevent cyberattacks in real-time.

## 5. Facilitating Decentralized Coordination

- In-network computing enables **peer-to-peer (P2P) communication** between drones and ground infrastructure without requiring all interactions to go through a central server.
- This is particularly useful for **decentralized UTM approaches**, where different stakeholders (e.g., ANSPs, drone service providers) need to dynamically share information.

## 6. Reducing Dependency on Cloud Connectivity

- Traditional UTM systems rely on cloud-based services, which can introduce **single points of failure**.
- In-network computing enables **local failover mechanisms**, ensuring continued operations even in case of network disruptions.

## 7. Enabling AI-Driven Adaptive Traffic Management

- Edge nodes can integrate **AI and machine learning models** to predict traffic patterns and adjust drone routing dynamically.
- This proactive management reduces conflicts and improves airspace efficiency.

## Conclusion

In-network computing aligns well with the vision of **distributed, scalable, and real-time communication** for U-Space UTM systems. It enhances **latency reduction, security, scalability, and autonomous decision-making**, making it a critical enabler for the **next-generation drone traffic management**.

Creating an **In-Network Computing (ICN)**-focused **simulation test setup** requires designing a testbed that integrates **ICN principles** with a **network simulation environment**. Below is a structured approach to setting up such a simulation,

specifically tailored for applications like **Unmanned Traffic Management (UTM) in U-Space**.

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## 1. Define Objectives for ICN Simulation

Before setting up the simulation, clarify the goals:

- **Test distributed computing in the network fabric** (instead of centralized cloud-based processing).
  - **Evaluate the efficiency of ICN in reducing network congestion** for UTM-like applications.
  - **Analyze the performance of Named Data Networking (NDN)** in drone-to-drone (D2D) and drone-to-ground (D2G) communication.
  - **Optimize real-time decision-making for U-Space** through edge-based computation.
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## 2. Select a Simulation Framework

### Network Simulators with ICN Support

Choose a network simulator that supports ICN concepts such as **NDN (Named Data Networking)** or **CCN (Content-Centric Networking)**:

- **ndnSIM** (NDN Simulator for NS-3) → Best for ICN-based networking research.
  - **NS-3** with ICN extensions → Useful for performance testing in software-defined networking.
  - **Mini-ICN** (ICN for Mininet) → Good for emulating in-network computing on software-defined networks.
  - **OMNeT++** with INET/ICN extensions → For detailed protocol analysis in complex network environments.
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## 3. Setup Simulation Environment

### A. Install Required Tools

Depending on the simulator, set up the necessary environment:

- **For ndnSIM (on NS-3):**

```
bash
CopyEdit
git clone https://github.com/named-data-ndnSIM/ns-3-dev-ndnSIM.git
cd ns-3-dev-ndnSIM
./waf configure
./waf build
```

- **For Mini-ICN:**

```
bash
CopyEdit
git clone https://github.com/ICN-SDN/Mini-ICN.git
cd Mini-ICN
sudo ./setup.sh
```

- **For OMNeT++ with ICN support:**

```
bash
CopyEdit
git clone https://github.com/inet-framework/inet.git
```

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## 4. Define Network Topology

Model the UTM/U-Space communication structure within the simulation:

- **Nodes (Drones, Edge Nodes, Ground Stations)**
- **Network Infrastructure (5G/6G, SDN, Edge Servers)**
- **ICN Routers (NDN-enabled nodes for in-network caching and computation)**

For **ndnSIM**, create a topology file:

```

cpp
CopyEdit
// Define a simple topology with 3 drones, 2 edge nodes, and a ground control station
NodeContainer drones, edgeNodes, groundStation;
drones.Create(3);
edgeNodes.Create(2);
groundStation.Create(1);

// Connect drones to edge nodes
PointToPointHelper p2p;
p2p.Install(drones.Get(0), edgeNodes.Get(0));
p2p.Install(drones.Get(1), edgeNodes.Get(1));
p2p.Install(drones.Get(2), edgeNodes.Get(0));

// Connect edge nodes to ground station
p2p.Install(edgeNodes.Get(0), groundStation.Get(0));
p2p.Install(edgeNodes.Get(1), groundStation.Get(0));

```

## 5. Implement ICN-Based In-Network Computing

- Deploy an **NDN Forwarding Strategy** that enables in-network caching and computing.
- Modify the **ndnSIM forwarding strategy** to process data within the network instead of relaying to a central cloud.

### Example: Adding an Edge Compute Function

Modify `ndn-forwarding-strategy.cpp` to include an in-network computing function:

```

cpp
CopyEdit
if (interest.matches("/utm/data")) {
    // Process data locally at the edge instead of forwarding
    std::string processedData = computeFunction(interest.getData());
    sendToConsumer(processedData);
}

```

```
}
```

## 6. Simulate Drone Communications in UTM Context

### A. Define UTM Data Flows

- **Drone-to-Drone (D2D) ICN Messaging** (e.g., sharing telemetry data)
- **Drone-to-Ground Control (D2G) Communication** (e.g., requesting navigation assistance)
- **Edge Processing for Conflict Detection** (e.g., computing alternative flight paths in-network)

### B. Simulate ICN Content Caching and Processing

- Implement **NDN-based caching** to reduce redundant data transmission.
- Deploy an **edge-based decision-making algorithm** for UTM.

Example: Drone requests weather data from the network:

```
cpp
CopyEdit
auto interest = ndn::Interest("/utm/weather");
interest.setMustBeFresh(true);
m_face.expressInterest(interest, onData, onTimeout);
```

## 7. Analyze Performance Metrics

Use built-in logging and packet tracing in the simulator to evaluate:

- **End-to-end latency reduction**
- **Bandwidth utilization efficiency**
- **Processing time of edge-computed tasks**
- **Cache hit ratio for repeated requests**

Example: Logging delay statistics in `ndnSIM`:

```
cpp
CopyEdit
NS_LOG_INFO("Average Delay: " << delaySum / totalRequests);
```

## 8. Extend to Real-World Testbeds

Once simulation results are promising:

- Deploy on **testbeds like NDN Testbed, CloudLab, or Fed4FIRE**.
- Implement real-world **ICN routers (e.g., Cisco NDN routers)**.

## Conclusion

This setup provides a **fully simulated ICN-based in-network computing testbed** for **UTM in U-Space**. It leverages **ICN caching, distributed processing, and efficient data retrieval** to optimize drone communication. Future work can involve integrating **AI-driven edge computing** for even more autonomous decision-making.

While in-network computing has been explored in various domains, its application specifically to Unmanned Aircraft System Traffic Management (UTM) and Advanced Air Mobility (AAM) is still emerging. Current research primarily focuses on integrating advanced air mobility operations with existing mobile networks, particularly 5G and future 6G systems, to meet the stringent requirements of AAM use cases. For instance, studies have examined how 5G systems can support AAM operations by analyzing key features, benefits, and potential gaps in leveraging mobile networks for such applications. [arc.aiaa.org](https://arc.aiaa.org)+[en.wikipedia.org](https://en.wikipedia.org)+[faa.gov](https://www.faa.gov)+[fda.gov](https://www.fda.gov)

In-network computing, which involves processing data within the network infrastructure to reduce latency and improve efficiency, has been applied in areas like machine learning and data transmission optimization. For example, frameworks have been developed to map machine learning models onto programmable network devices, enabling real-time data processing within the network.

Additionally, network transmission paradigms utilizing in-network computing have been proposed to enhance data transmission and computation processing collaboratively . [changgang-zheng.github.io](https://github.com/changgang-zheng)

However, the direct application of in-network computing to UTM and AAM systems remains an area with limited research. As AAM operations are expected to demand low-latency, high-reliability communications, integrating in-network computing could potentially enhance the performance of UTM systems by enabling real-time data processing and decision-making within the network. This integration could support functions such as dynamic route planning, collision avoidance, and traffic management for unmanned aircraft.

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Given the nascent stage of this research intersection, there is a significant opportunity for further exploration into how in-network computing can be tailored to meet the specific needs of UTM and AAM applications. Future studies could focus on developing architectures that incorporate in-network processing capabilities into UTM frameworks, evaluating the performance benefits, and addressing challenges related to scalability, security, and interoperability with existing air traffic management systems.[pmc.ncbi.nlm.nih.gov](https://pmc.ncbi.nlm.nih.gov)

In-network computing (INC) is an emerging paradigm that enhances data transmission and processing by performing computations within the network infrastructure itself. This approach leverages programmable network devices, such as switches and routers, to execute tasks traditionally handled by end-hosts or dedicated servers, thereby reducing latency, optimizing bandwidth usage, and improving overall system performance.

### **Key Concepts of In-Network Computing:**

1. **Programmable Data Planes:** Modern network devices can be programmed to process data packets on-the-fly. This capability allows for the offloading of specific computational tasks to the network hardware, enabling real-time data processing and reducing the need to transmit data back and forth between endpoints.
2. **Collaborative Data Transmission and Computation:** INC facilitates a collaborative approach where data transmission and computation are intertwined. By processing data as it traverses the network, **INC reduces the volume of data transmitted and accelerates response times.**

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3. **Enhanced Network Efficiency:** By distributing computational tasks across network devices, INC alleviates bottlenecks associated with centralized processing systems. This distribution leads to more efficient use of network resources and improved scalability.

### **Notable Research in In-Network Computing:**

1. **"A Survey on In-Network Computing: Programmable Data Plane and Technology Specific Applications"**

*Authors:* J. Song, Z. Li, and Y. Cheng

*Summary:* This comprehensive survey discusses various use cases, enabling technologies, and protocols associated with in-network computing. It emphasizes the role of programmable data planes in facilitating INC and explores applications across different technological domains.

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2. **"In-Network Computing Based Transmission Optimization Mechanism"**

*Authors:* Y. Li, Y. Zhang, and D. Raychaudhuri

*Summary:* This paper proposes a network transmission paradigm that combines data transmission with computation processing through in-network computing. The approach aims to optimize network performance by leveraging the computational capabilities of network devices.

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### 3. **"Process-in-Network: A Comprehensive Network Processing Approach"**

*Authors:* J. Lloret, M. Garcia, and S. Sendra

*Summary:* The article introduces the concept of Process-in-Network (PIN), where the network processes information during transmission. This approach utilizes idle processing capacities in intermediate nodes, enhancing the efficiency of data handling within the network. [pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov)

### 4. **"P4COM: In-Network Computation with Programmable Switches"**

*Authors:* G. Chen, G. Zeng, and L. Chen

*Summary:* This paper presents P4COM, a framework that automates the mapping of machine learning models onto programmable network devices. By enabling in-network machine learning, P4COM demonstrates the potential of INC in handling data-intensive tasks directly within the network infrastructure. [arxiv.org](https://arxiv.org)

These studies collectively highlight the transformative potential of in-network computing in modern data transmission and processing scenarios. By integrating computational capabilities into the network fabric, INC offers a promising avenue for enhancing the performance and efficiency of distributed systems.

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## Implementing In-Network Computing (INC) for UTM (Unmanned Aircraft System Traffic Management) and U-space (EU's drone traffic management framework)

requires a systematic approach, integrating network programmability, edge computing, and AI-driven decision-making within the air traffic infrastructure. Here's a structured roadmap to begin:

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### 1. Define Key Objectives & Use Cases

Before implementation, define the critical applications where INC can provide value:

- **Real-time Conflict Detection & Avoidance:** Process telemetry data mid-network to detect potential conflicts earlier.

- **Dynamic Route Optimization:** Compute alternative flight paths within the network, reducing cloud processing delays.
  - **Federated U-space Data Aggregation:** Distribute air traffic data processing across multiple network nodes for seamless operations.
  - **Edge-based Command & Control (C2) Execution:** Move UAS control logic closer to the network edge for ultra-low latency.
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## 2. Select Suitable Network Infrastructure

- **Adopt Software-Defined Networking (SDN):** Enables centralized control over traffic management.
  - **Use Programmable Data Planes (P4, eBPF, XDP):** Implements custom in-network computation logic in network switches and routers.
  - **Integrate 5G/6G Networks:** Enables MEC (Multi-access Edge Computing) to offload U-space services.
  - **Leverage Satellite & V2X Communication:** Ensures high connectivity for BVLOS (Beyond Visual Line of Sight) operations.
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## 3. Develop In-Network Processing Pipelines

- **Telemetry & Sensor Fusion:** Aggregate ADS-B, Remote ID, and flight telemetry data at the network level.
  - **AI-Driven Decision Making:** Implement lightweight ML models for anomaly detection directly in network switches.
  - **Distributed Data Processing:** Use in-network compute resources to preprocess UAV telemetry before sending it to the cloud.
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## 4. Implement Prototype Using Available Tools

- **Use P4-enabled Switches (e.g., Tofino, BMv2):** Program in-network packet processing for real-time drone data handling.
- **Leverage eBPF/XDP for U-space Edge Processing:** Deploy security and anomaly detection in Linux-based network nodes.
- **Integrate MEC for U-space Data Offloading:** Utilize 5G edge computing to support real-time UAV operations.

- **Develop U-space APIs using gRPC & MQTT:** Enable efficient data exchange between UTM, drones, and network nodes.
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## 5. Test & Optimize the System

- **Simulate in a Digital Twin Environment:** Use **AirSim, Gazebo, or BlueSky** for U-space air traffic emulation.
  - **Deploy in a Controlled UAS Test Corridor:** Validate in-network processing effectiveness with real drones.
  - **Optimize for Latency & Reliability:** Measure real-time performance gains versus traditional cloud processing.
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## 6. Scale to Real-World Deployment

- **Collaborate with ANSPs (Air Navigation Service Providers):** Ensure integration with national airspace infrastructure.
- **Standardize with U-space & ASTM Standards:** Align with protocols like **EU U-space Regulation, ASTM F3411 (Remote ID), and EUROCAE WG-105.**
- **Implement Failover Mechanisms:** Ensure high availability and security with distributed redundancy.

Here's a **high-level architecture** for an **In-Network Computing (INC)-based UTM (Unmanned Aircraft System Traffic Management) system**, integrating **U-space services** and edge processing capabilities.

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## INC-Based UTM Architecture

### Key Components:

#### 1. Drone (UAS) Layer 🚁

- UAVs with onboard **ADS-B, Remote ID, and C2 Links**
- 5G/6G connectivity with **edge computing-enabled UAV radios**
- AI-based autonomous navigation

#### 2. Edge Network & In-Network Computing Layer 🌐

- **5G Multi-access Edge Computing (MEC) Nodes:** Low-latency, in-network computations
  - **Programmable Switches (P4/eBPF/XDP):** In-network packet processing for telemetry, security, and routing
  - **AI/ML-Based Traffic Analysis & Prediction:** Real-time conflict detection in switches
  - **V2X Communication for Direct Drone-to-Drone Coordination**
3. **UTM & U-space Traffic Management Layer** 
- **SDN Controller:** Dynamic air traffic flow control via network-based policies
  - **Geo-Fencing & No-Fly Zone Enforcement:** Distributed policy enforcement within the network
  - **Real-Time Airspace Congestion Monitoring**
  - **Collision Avoidance Using AI Models in the Network**
4. **Cloud & Data Services Layer** 
- **Centralized UTM Coordination Server (NASA USS, U-space Services)**
  - **Federated Data Aggregation (Cross-Network Processing)**
  - **Digital Twin Integration (Simulation & Training Environments)**
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## **System Workflow:**

1. **UAV Telemetry Processing in the Network**
  - UAVs transmit real-time data (position, speed, altitude) via **Remote ID & ADS-B**
  - Network **preprocesses telemetry using INC** before sending to UTM servers
2. **Dynamic Airspace Management**
  - **SDN policies optimize air traffic routing dynamically**
  - Real-time conflict detection via **AI inference at edge switches**
3. **Low-Latency Command & Control**
  - MEC enables **in-network execution of UTM commands**
  - **Real-time re-routing in case of congestion/conflict**

#### 4. Security & Compliance Enforcement

- In-network anomaly detection identifies **rogue drones**
  - **Distributed geo-fencing policies prevent unauthorized entry**
- 

### Technologies to Implement

- ✓ **Programmable Networks: P4, eBPF, XDP**
  - ✓ **5G/6G MEC for Edge Computing**
  - ✓ **AI/ML Models for Traffic Prediction & Collision Avoidance**
  - ✓ **SDN-Based UTM Traffic Control**
  - ✓ **Secure Federated Learning for Cross-Network U-space Coordination**
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### Next Steps

Would you like a **detailed implementation plan**, or help with **coding specific in-network functions (e.g., P4 for telemetry processing, AI for traffic prediction, or SDN for routing control)**?

INC - for conflict detection in switches - Mesh